Global Retail Data Audit & Optimization (113k+ Records)

Role: Data Analyst

Objective: To standardize and audit a massive retail dataset to ensure financial reporting accuracy and identify profit-driving insights.

1. The Data Integrity Challenge

The raw dataset contained over **113,000 rows** of global sales records but suffered from significant "data noise" that prevented accurate analysis:

- **Inconsistent Date Formats:** Mixed separators and typos (e.g., "Fanuary") that blocked time-series trends.
- **Text String Errors:** "Dirty" data in the revenue and profit columns that prevented mathematical calculations.
- **Scale Complexity:** The volume of data made manual correction impossible, requiring an automated cleaning approach.

Raw Dataset:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_Age	Age_Group	Customer_Gender	Country	State	Product_Category	Sub_Category	Product	Order_Quantity	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	26/11/2013	26	November	2013	19	Youth (<25)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	8	45	120	590	360	950
3	26/11/2015	26	November	2015	19	Youth (<25)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	8	45	120	590	360	950
4	23/03/2014	23	March	2014	49	Adults (35-64)	M	Australia	New Soutl	Accessories	Bike Racks	Hitch Rack	23	45	120	1366	1035	2401
5	23/03/2016	23	March	2016	49	Adults (35-64)	M	Australia	New Soutl	Accessories	Bike Racks	Hitch Rack	20	45	120	1188	900	2088
6	15/05/2014	15	May	2014	47	Adults (35-64)	F	Australia	New Soutl	Accessories	Bike Racks	Hitch Rack	4	45	120	238	180	418
7	15/05/2016	15	May	2016	47	Adults (35-64)	F	Australia	New Soutl	Accessories	Bike Racks	Hitch Rack	5	45	120	297	225	522
8	22/05/2014	22	May	2014	47	Adults (35-64)	F	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	4	45	120	199	180	379
9	22/05/2016	22	May	2016	47	Adults (35-64)	F	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	2	45	120	100	90	190
10	22/02/2014	22	February	2014	35	Adults (35-64)	M	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	22	45	120	1096	990	2086
11	22/02/2016	22	February	2016	35	Adults (35-64)	M	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	21	45	120	1046	945	1991
12	30/07/2013	30	July	2013	32	Young Adults (25-34)	F	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	8	45	120	398	360	758
13	30/07/2015	30	July	2015	32	Young Adults (25-34)	F	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	8	45	120	398	360	758
14	15/07/2013	15	July	2013	34	Young Adults (25-34)	M	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	7	45	120	349	315	664
15	15/07/2015	15	July	2015	34	Young Adults (25-34)	M	Australia	Victoria	Accessories	Bike Racks	Hitch Rack	7	45	120	349	315	664
16	02/08/2013	2	August	2013	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	5	45	120	369	225	594
17	02/08/2015	2	August	2015	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	7	45	120	517	315	832
18	02/09/2013	2	Septembe	2013	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	2	45	120	148	90	238
19	02/09/2015	2	Septembe	2015	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	1	45	120	74	45	119
20	22/01/2014	22	January	2014	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	1	45	120	74	45	119
21	22/01/2016	22	January	2016	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	1	45	120	74	45	119
22	17/05/2014	17	May	2014	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	6	45	120	443	270	713
23	17/05/2016	17	May	2016	29	Young Adults (25-34)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	8	45	120	590	360	950
24	27/03/2014	27	March	2014	51	Adults (35-64)	M	United Stat	Oregon	Accessories	Bike Racks	Hitch Rack	9	45	120	524	405	929
25	27/03/2016	27	March	2016	51	Adults (35-64)	M	United Stat	Oregon	Accessories	Bike Racks	Hitch Rack	7	45	120	407	315	722
26	25/08/2013	25	August	2013	49	Adults (35-64)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	3	45	120	221	135	356
27	25/08/2015	25	August	2015	49	Adults (35-64)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	1	45	120	74	45	119
28	26/12/2013	26	December	2013	49	Adults (35-64)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	6	45	120	443	270	713
29	26/12/2015	26	December	2015	49	Adults (35-64)	M	Canada	British Col	Accessories	Bike Racks	Hitch Rack	5	45	120	369	225	594
30	02/01/2014	2	January	2014	48	Adults (35-64)	F	Canada	British Col	Accessories	Bike Racks	Hitch Rack	4	45	120	295	180	475
31	02/01/2016	2	January	2016	48	Adults (35-64)	F	Canada	British Col	Accessories	Bike Racks	Hitch Rack	4	45	120	295	180	475
32	13/03/2014	13	March	2014	48	Adults (35-64)	F	Canada	British Col	Accessories	Bike Racks	Hitch Rack	20	45	120	1476	900	2376
33	13/03/2016	13	March	2016	48	Adults (35-64)	F	Canada	British Co	Accessories	Bike Racks	Hitch Rack	22	45	120	1624	990	2614
34	23/05/2014	23	May	2014	47	Adults (35-64)	F	Canada	British Col	Accessories	Bike Racks	Hitch Rack	24	45	120	1771	1080	2851
35	23/05/2016	23	May	2016	47	Adults (35-64)	F	Canada	British Col	Accessories	Bike Racks	Hitch Rack	23	45	120	1697	1035	2732

2. The Transformation Process

I implemented a systematic cleaning pipeline to transform the raw data into a "BI-Ready" asset:

- **Step 1: Normalization:** Standardized all categorical fields (Product, Country, Gender) to ensure uniform grouping.
- **Step 2: Financial Audit:** Scrubbed non-numeric characters from the Revenue and Profit columns, casting them into precise decimal formats.
- **Step 3: Logical Verification:** Validated that **Revenue - Cost = Profit** across every record to ensure no data corruption occurred during the process.

Power Query:

```
= Table.RenameColumns(#"Changed Type4",{{"Customer_Age", "Customer Age"}, {"Age_Group", "Age Group"}, {"Customer_Gender", "Customer Gender"}, {"Sub_Category", "Sub Category"}, {"Unit_Cost", "Unit Cost"},
```

	Date	Customer Age	Age Group	Customer Gender	Country	State	Product Category	Sub Category	Product
	Valid 100% Error 0% Empty 0% 556 distinct, 274 unique	Valid 100% Error 0% Empty 0% 52 distinct, 0 unique	Valid 100% Error 0% Empty 0% 4 distinct, 0 unique	Valid 100% Error 0% Empty 0% 2 distinct, 0 unique	Valid 100% Error 0% Empty 0% 6 distinct, 0 unique	Valid 100% Error 0% Empty 0% 29 distinct, 0 unique	Valid 100% Error 0% Empty 0% 1 distinct, 0 unique	Valid 100% Error 0% Empty 0% 2 distinct, 0 unique	Valid 100% Error 0% Empty 0% 2 distinct, 0 unique
1	26/11/2013	19	Youth (<25)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
2	26/11/2015	19	Youth (<25)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
3	23/03/2014	49	Adults (35-64)	Male	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike
4	23/03/2016	49	Adults (35-64)	Male	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike
5	15/05/2014	47	Adults (35-64)	Female	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike
6	15/05/2016	47	Adults (35-64)	Female	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike
7	22/05/2014	47	Adults (35-64)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
8	22/05/2016	47	Adults (35-64)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
9	22/02/2014	35	Adults (35-64)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
10	22/02/2016	35	Adults (35-64)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
11	30/07/2013	32	Young Adults (25-34)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
12	30/07/2015	32	Young Adults (25-34)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
13	15/07/2013	34	Young Adults (25-34)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
14	15/07/2015	34	Young Adults (25-34)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike
15	02/08/2013	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
16	02/08/2015	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
17	02/09/2013	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
18	02/09/2015	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
19	22/01/2014	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
20	22/01/2016	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
21	17/05/2014	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike
22	17/05/2016	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike

Cleaned dataset:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Customer Age	Age Group	Customer Gender	Country	State	Product Category	Sub Category	Product	Order Quantity	Unit Cost	Unit Price	Profit	Loss	Cost	Revenue	%Profit	%Loss
2	26/11/2013	19	Youth (<25)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45	120	£590.00	£240.00	£360.00	£950.00	62%	67%
3	26/11/2015	19	Youth (<25)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45	120	£590.00	£240.00	£360.00	£950.00	62%	67%
4	23/03/2014	49	Adults (35-64)	Male	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	23	45	120	£1,366.00	£915.00	£1,035.00	£2,401.00	57%	88%
5	23/03/2016	49	Adults (35-64)	Male	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	20	45	120	£1,188.00	£780.00	£900.00	£2,088.00	57%	87%
6	15/05/2014	47	Adults (35-64)	Female	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	4	45	120	£238.00	£60.00	£180.00	£418.00	57%	33%
7	15/05/2016	47	Adults (35-64)	Female	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	5	45	120	£297.00	£105.00	£225.00	£522.00	57%	47%
8	22/05/2014	47	Adults (35-64)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	4	45	120	£199.00	£60.00	£180.00	£379.00	53%	33%
9	22/05/2016	47	Adults (35-64)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	2	45	120	£100.00	£0.00	£90.00	£190.00	53%	0%
10	22/02/2014	35	Adults (35-64)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	22	45	120	£1,096.00	£870.00	£990.00	£2,086.00	53%	88%
11	22/02/2016	35	Adults (35-64)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	21	45	120	£1,046.00	£825.00	£945.00	£1,991.00	53%	87%
12	30/07/2013	32	Young Adults (25-34)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45	120	£398.00	£240.00	£360.00	£758.00	53%	67%
13	30/07/2015	32	Young Adults (25-34)	Female	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45	120	£398.00	£240.00	£360.00	£758.00	53%	67%
14	15/07/2013	34	Young Adults (25-34)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	7	45	120	£349.00	£195.00	£315.00	£664.00	53%	62%
15	15/07/2015	34	Young Adults (25-34)	Male	Australia	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	7	45	120	£349.00	£195.00	£315.00	£664.00	53%	62%
16	02/08/2013	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	5	45	120	£369.00	£105.00	£225.00	£594.00	62%	47%
17	02/08/2015	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	7	45	120	£517.00	£195.00	£315.00	£832.00	62%	62%
18	02/09/2013	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	2	45	120	£148.00	£0.00	£90.00	£238.00	62%	0%
19	02/09/2015	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	1	45	120	£74.00	£0.00	£45.00	£119.00	62%	0%
20	22/01/2014	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	1	45	120	£74.00	£0.00	£45.00	£119.00	62%	0%
21	22/01/2016	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	1	45	120	£74.00	£0.00	£45.00	£119.00	62%	0%
22	17/05/2014	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	6	45	120	£443.00	£150.00	£270.00	£713.00	62%	56%
23	17/05/2016	29	Young Adults (25-34)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45	120	£590.00	£240.00	£360.00	£950.00	62%	67%
24	27/03/2014	51	Adults (35-64)	Male	United States	Oregon	Accessories	Bike Racks	Hitch Rack - 4-Bike	9	45	120	£524.00	£285.00	£405.00	£929.00	56%	70%
25	27/03/2016	51	Adults (35-64)	Male	United States	Oregon	Accessories	Bike Racks	Hitch Rack - 4-Bike	7	45	120	£407.00	£195.00	£315.00	£722.00	56%	62%
26	25/08/2013	49	Adults (35-64)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	3	45	120	£221.00	£15.00	£135.00	£356.00	62%	11%
27	25/08/2015	49	Adults (35-64)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	1	45	120	£74.00	£0.00	£45.00	£119.00	62%	0%
28	26/12/2013	49	Adults (35-64)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	6	45	120	£443.00	£150.00	£270.00	£713.00	62%	56%
29	26/12/2015	49	Adults (35-64)	Male	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	5	45	120	£369.00	£105.00	£225.00	£594.00	62%	47%
30	02/01/2014	48	Adults (35-64)	Female	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	4	45	120	£295.00	£60.00	£180.00	£475.00	62%	33%
31	02/01/2016	48	Adults (35-64)	Female	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	4	45	120	£295.00	£60.00	£180.00	£475.00	62%	33%
32	13/03/2014	48	Adults (35-64)	Female	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	20	45	120	£1,476.00	£780.00	£900.00	£2,376.00	62%	87%

3. Executive Results & Insights

After cleaning the data, I distilled the 113,000+ rows into an **Executive Summary** that revealed the true health of the business:

- **Total Revenue: $85.27M**.
- **Total Profit: $32.22M**.
- **Profit Margin:** A strong **37.8%** overall.
- **Top Performance:** Identified "Bikes" as the primary revenue driver ($61.7M).
- **Operational Risk:** Pinpointed specific loss-making products where costs exceeded revenue.